

The Persistence Ceiling: Testing Whether Form, Tournaments, and Squad Value Improve National-Team Elo Forecasts

Soccer

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Abstract.

National-team federations routinely read recent form, tournament performance, volatility, and squad value as signs that a team may be rising or fading. These signals, however, do not necessarily tell the same story. Some describe where a team stands today, some may contain information about the direction of future movement, and only some may translate into lower forecast error. This study separates those roles by testing the additional forecasting information carried by richer historical signals when current Elo is already known.

We use a frozen national-team-season modelling panel covering rated men's national teams from 1980-2026, with 10,301 team-season observations at the one-step horizon and future Elo change evaluated through $t+4$. Future Elo is the forecasting outcome rather than a claim of ground-truth football quality. Models are evaluated with expanding-window, rolling-origin, out-of-time pooling. Persistence provides the reference benchmark, while progressively richer alternatives incorporate Elo level, trend, tournament history, volatility scaling, mean reversion, trend-plus-volatility, and team-specific change scale. MAE measures forecast accuracy, while Pearson and Spearman correlations are used only to assess directional signal. Feature availability changes the number of evaluable rows for some specifications, so raw MAE is reported without treating unmatched comparisons as paired effects.

Persistence records MAE of 28.62, 41.49, 49.62, and 56.56 from $t+1$ through $t+4$, and none of the tested feature families produces lower out-of-time MAE in this evaluation. This result is interpreted cautiously. Because Elo itself updates in response to match-result surprises, carrying the current rating forward is naturally a demanding benchmark when future Elo is the target. The more informative pattern is the separation between direction and error. Trend-plus-volatility raises directional correlation from 0.073 to 0.134 across horizons while still failing to improve MAE. Figure 1 illustrates this distinction: additional signal about direction does not automatically translate into a more accurate forecast of the future rating level.

Figure 1. Persistence Benchmark and Direction-versus-Error Robustness

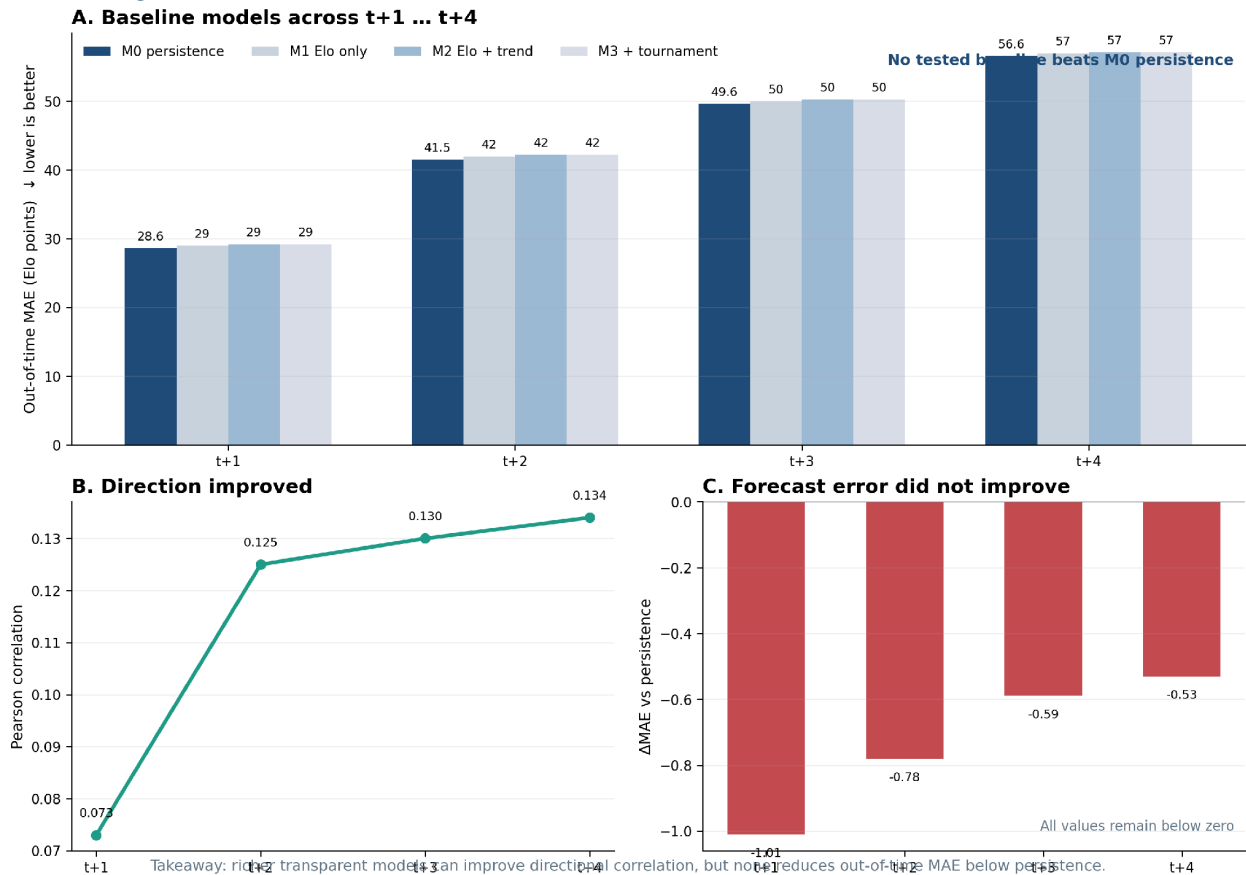


Figure 1. Persistence benchmark and direction-versus-error robustness: richer models can improve directional correlation while still failing to beat persistence on out-of-time MAE.

Squad value provides a complementary view of the same distinction. In the 112-team snapshot, log10 squad market value is strongly associated with current Elo, with Pearson correlation 0.911 and Spearman correlation 0.924; the corresponding raw-EUR Pearson correlation is 0.657. The available t+1 value exercise is an in-sample snapshot diagnostic rather than a historical rolling-origin out-of-time test. Within that diagnostic, Elo-plus-value records MAE of 26.14 compared with 26.28 for the fitted Elo-only model, while persistence is 26.09. The snapshot therefore supports market value as useful context for current team strength, but it does not establish an out-of-time forecasting gain. Figure 2 reinforces the broader distinction between describing current level and anticipating future movement. Taken together, the results show why federation analysis benefits from separating three questions: what describes current level, what contains directional information, and what demonstrably reduces forecast error. The findings are predictive rather than causal and remain conditional on the tested models, horizons, feature availability, and snapshot-only valuation coverage.

Figure 2. LEVEL $\neq$ CHANGE: Squad Market Value Describes Current Strength Better Than Future Movement

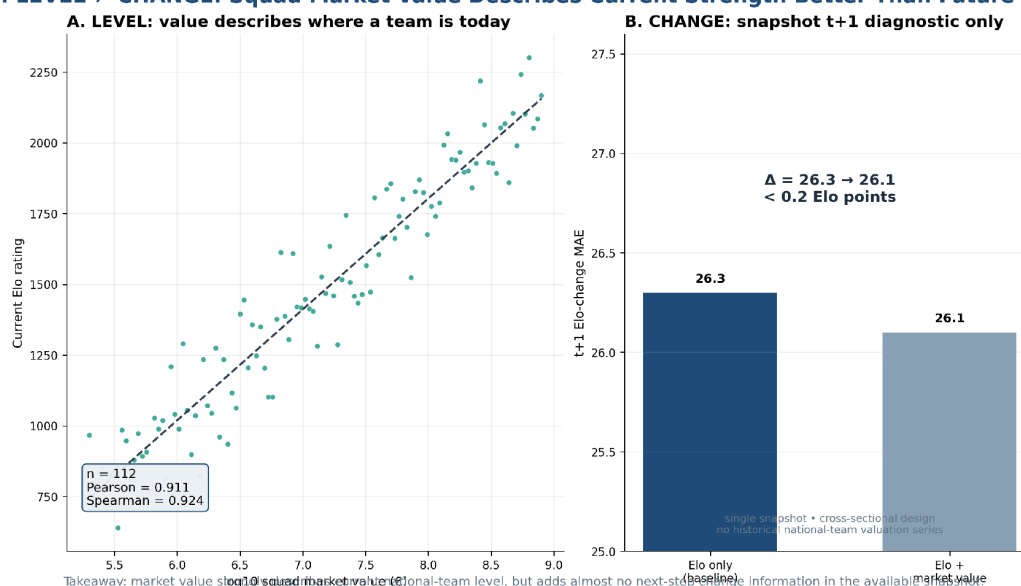


Figure 2. LEVEL $\neq$ CHANGE: squad market value strongly describes current strength, but adds almost no next-step change information in the available snapshot.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: [PUBLIC ANONYMIZED GITHUB URL REQUIRED BEFORE SUBMISSION]